

Here's your complete final detail — everything in one place 📌

ExamShield

Real-Time Secure Online Examination & Proctoring System

Project Definition

ExamShield ek web-based secure online examination platform chhe jema Admin question bank manage kare chhe, Student ne seed-based shuffle thi automatically unique question set assign thay, ane Proctor real-time ma cheating attempts monitor kare chhe. System tab switch, copy-paste, window blur jeva cheating attempts browser na native JavaScript events thi detect kare chhe ane WebSocket thi proctor ne instantly alert aapay chhe. Page reload karo to pan same questions aave chhe — kyunki question sequence database ma save thay chhe. Exam pura thayta MCQ ane True/False auto-grade thay chhe ane result sathe rank list automatically generate thay chhe.

Roles & Responsibilities

Admin

- Question bank manage kare — manually add kare ya Excel upload thi bulk import
- Exam create kare — title, duration, date, passing marks, total questions
- Decide kare — 100 questions bank ma chhe, student ne 30 aapva chhe
- Difficulty ratio set kare — 10 Easy + 15 Medium + 5 Hard
- Proctor assign kare — kon si exam monitor karse
- Results ane analytics jue — toppers, average score, pass/fail ratio
- Rules configure kare — ketla violations pachi student suspend thay

Proctor

- Live dashboard — badha active students real-time ma jue
- Violation alert feed — tab switch, copy attempt instantly dikhe
- Student ne warn kari shake, exam suspend kari shake
- Exam pachi flagged students review kare ane marks approve kare
- Subjective questions manually grade kare

● Student

- Register/Login — secure auth
 - Available exams list jue
 - Exam start kare — unique shuffled 30 questions automatically aave
 - Timer chalu rahay — reload karo to pan same time remaining
 - Anti-cheat rules follow karva pade — tab switch, copy block
 - Submit kare — MCQ auto-grade
 - Result ane rank jue
 - Attempt history jue
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Shuffle Question Logic

Concept

Questions distribute nathi thata — drek student ne same 100 questions pool ma thi seed-based shuffle thi 30 questions select thay. Pool hamesha full rehay — questions finish nathi thata.

Seed Shu Chhe?

Seed ek unique number chhe. Same seed thi shuffle karo to hamesha same result aave. Drekk student no seed alag hoy chhe — isliye shuffle result alag.

Seed Kevi Rite Banay?

Seed = Student ID + Exam ID

Student 1, Exam 7 → Seed = 1_7

Student 2, Exam 7 → Seed = 2_7

Student 4, Exam 7 → Seed = 4_7

Shuffle Flow

100 Questions Pool — hamesha available (never finishes)

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Student exam start kare

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System seed banave — Student ID + Exam ID

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Seed thi 100 questions shuffle thay

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First 30 questions select thay

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Question sequence DB ma save thay

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Student ne 30 questions dikhe

Reload Problem

Student reload kare

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System DB ma check kare — attempt exists?

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YES → same saved sequence return kare → same 30 questions

NO → new attempt, new shuffle

Real Example

100 Questions Pool — Q1 to Q100

Student 1 (seed 1_7) → Q34, Q12, Q67, Q5, Q89 ... (30)

Student 2 (seed 2_7) → Q78, Q23, Q34, Q91, Q11 ... (30)

Student 3 (seed 3_7) → Q56, Q8, Q23, Q72, Q34 ... (30)

Student 4 (seed 4_7) → Q91, Q45, Q12, Q33, Q67 ... (30)

Student 1 reloads → same Q34, Q12, Q67, Q5, Q89 

500 students hoy → same logic, pool never finishes 

Overlap Hoy To Copy Thay?

Same question multiple students ne aavi shake — but order alag hoy chhe. Student 1 ne Q34 pehlo aavse, Student 4 ne Q34 15mun aavse. Copy impossible — kyunki kon question kyaay chhe te pata nathi.

? Question Types

Type 1 — MCQ

- 4 options — 1 correct
- Auto graded ☒

Type 2 — True / False

- 2 options only
- Auto graded ☒

Type 3 — Subjective / Short Answer

- Student text ma answer type kare
 - Manual grading — Proctor review kare ☒
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📦 Question Storage

Option 1 — Manual Add

Admin panel thi ek ek question add kare — question text, options, correct answer, topic, difficulty

Option 2 — Excel Bulk Upload

Admin Excel file upload kare — system Apache POI library thi parse kare ane database ma bulk insert kare

Database Ma Question Fields

Field	Example
Question Text	Java ma String immutable chhe?
Type	MCQ / TRUE_FALSE / SUBJECTIVE
Option A, B, C, D	True, False, Depends, None
Correct Answer	A

Field	Example
Topic	Java Core
Difficulty	EASY / MEDIUM / HARD
Marks	2

Security Features

Anti-Cheat — Browser Level

Feature	Technology	How
Tab Switch Detection	JavaScript — visibilitychange event	Browser detect kare tab change
Window Blur	JavaScript — window.blur event	ALT+TAB ya minimize detect
Copy Block	JavaScript — copy, cut events	Copy action prevent
Paste Block	JavaScript — paste event	Paste action prevent
Right Click Block	JavaScript — contextmenu event	Inspect element rokay

Koi external library nathi — pure browser native JavaScript events use karya chhe. Zero cost, zero dependency.

Violation Alert — WebSocket

Student tab switch kare

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JavaScript detect kare

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WebSocket thi Spring Boot server par message jay

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Server proctor dashboard par broadcast kare

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Proctor ne real-time alert — Name + Violation + Time

Auto Suspend

3 violations → exam automatically suspend

Proctor ne notification → review kare → decision leve

Timer — Redis

Timer server side Redis ma store chhe. Reload karo to frontend Redis thi remaining time fetch kare — reset nathi thatu.

Screen Recording — Honest Answer

Browser thi screen recording completely rokvu technically possible nathi — aa limitation Mettl, HackerEarth pan face kare chhe. Complete solution mate desktop application joiye. Amaro system web-based chhe — tab switch ane window blur detection college-level exam mate sufficient chhe.

Tech Stack

Technology	Purpose
Spring Boot	Backend — APIs, auth, exam logic, result generation
React	Frontend — exam UI, dashboards
PostgreSQL	Questions, students, attempts, violations, results store
Redis	Exam timer — server side, reload safe
WebSocket (STOMP)	Real-time violation alerts to proctor
JWT + Spring Security	Role-based login — Admin, Proctor, Student
Apache POI	Excel question bulk upload parse

Hard Questions & Answers

Q: Udemy jaisi existing platforms exist kare chhe — aa navu kyaan chhe?

"Udemy educational platform chhe — ExamShield examination ane proctoring system chhe. Banne alag domain chhe. Real problem solve kare chhe — college online exams ma cheating rokvu."

Q: Screen recording kevi rite rokso?

"Browser thi completely rokvu possible nathi — industry-wide same limitation chhe. Tab switch, window blur, copy-paste detection college-level mate sufficient chhe."

Q: 500 students same time exam de to server handle kare?

"WebSocket Spring Boot handle kare. Timer Redis ma chhe — database load nathi. College scale mate fully sufficient chhe."

Q: Same question multiple students ne aave to copy thay?

"Order alag hoy chhe — Student 1 ne Q34 pehlo aavse, Student 4 ne 15mun. Neighbor ne puchho to kon question kyaay chhe te pata nathi — copy impossible."

Q: Student mobile thi exam de to?

"System desktop-only chhe — same as professional proctoring tools. Mobile browser restricted message show kare."

Q: Reload kare to different questions aave?

"Na — question sequence DB ma save chhe. Reload karo to same sequence return thay. Reloading no koi faydo nathi."

Q: Subjective questions auto grade nathi thata?

"Correct — MCQ ane True/False auto-grade thay. Subjective manually proctor review kare ane marks enter kare — same as university paper checking."

 Why This Project is NOT Basic

Point	Explanation
Seed-based shuffle	Algorithmic thinking — not taught in college
WebSocket real-time	Not basic REST — live communication
Redis timer	Production pattern — server-side state
Role-based system	3 roles with different access levels
Anti-cheat JS	Frontend + Backend both covered
Excel bulk upload	Real admin workflow
Real problem	College khud use kari shake

Aa complete chhe — guide sathe confidently baat kari shake 🙌 Best of luck! 🔥